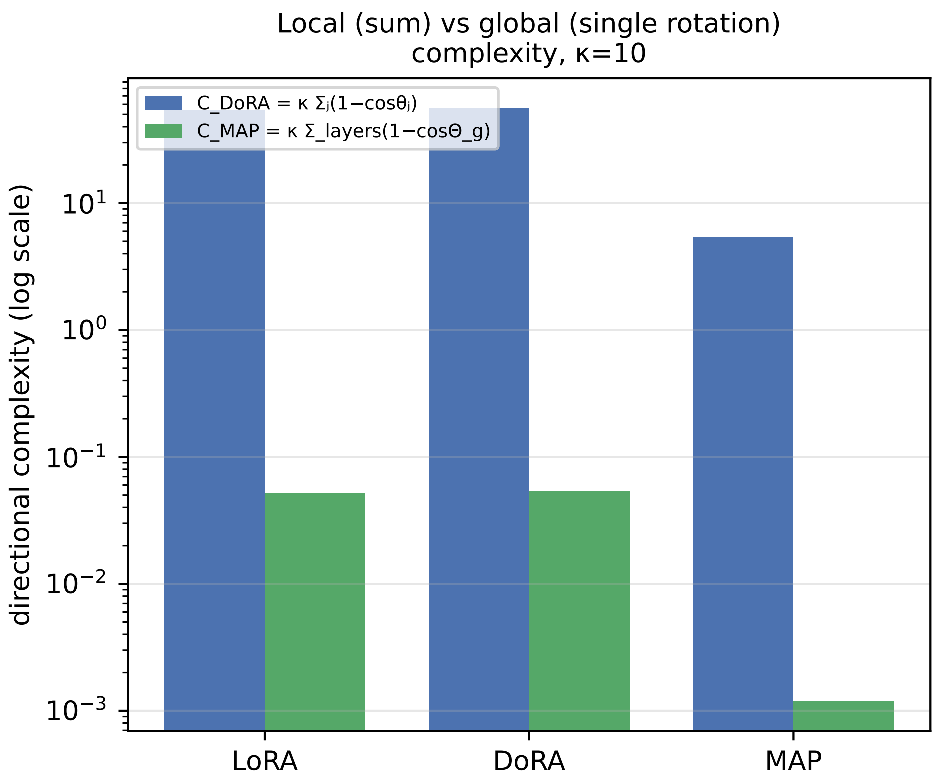
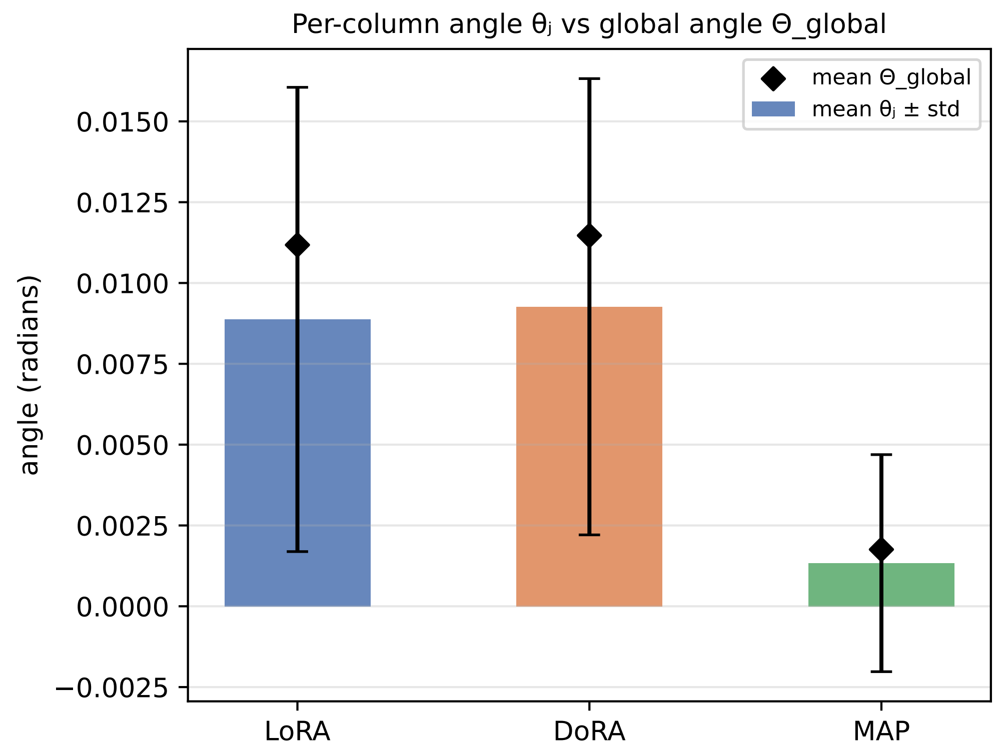
Directional Updates: LoRA vs DoRA vs MAP on RoBERTa-base / SST2

Fine-tuning metrics and aggregate directional complexity

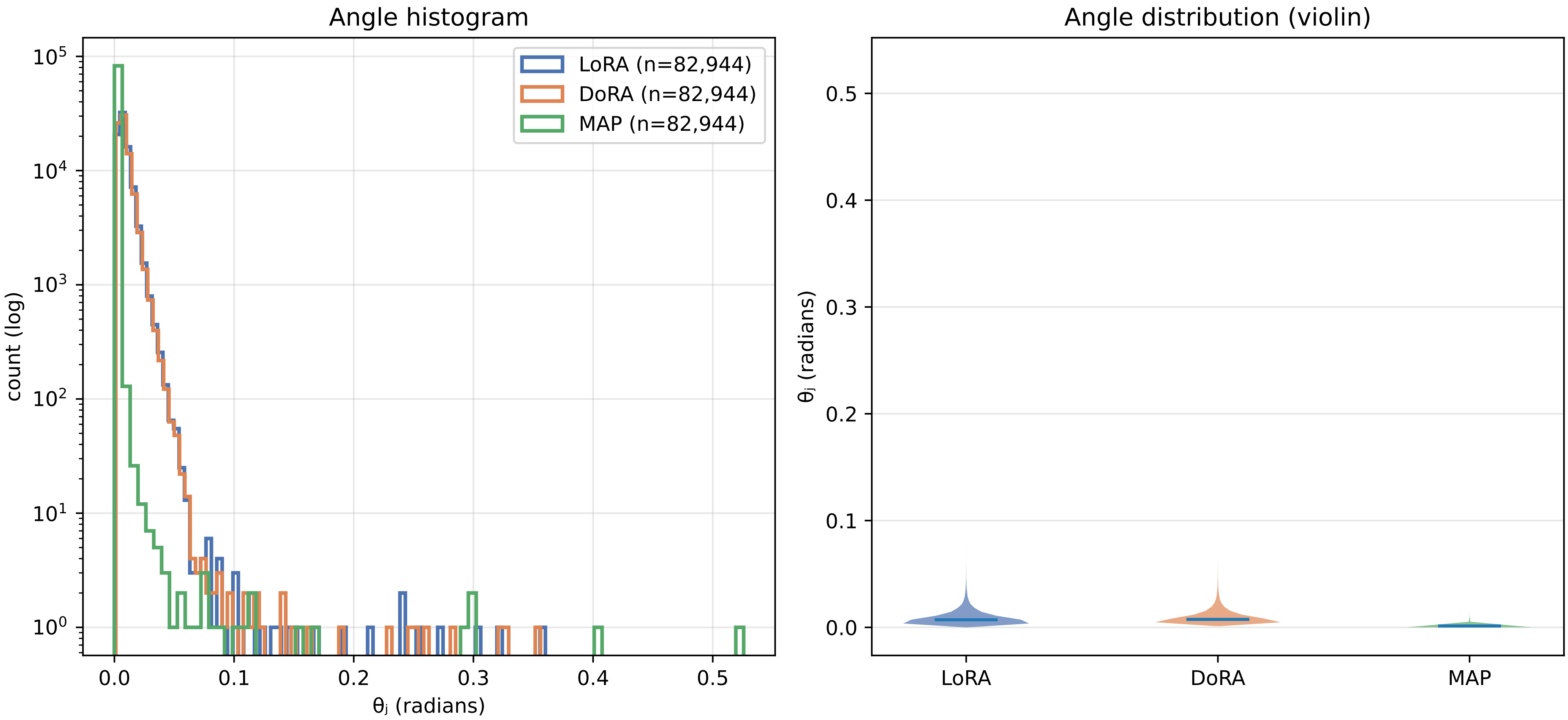
Method	Eval acc	Trainable params	C_DoRA (Σ local)	C_MAP (global)	mean θ_j (rad)	mean Θ_g (rad)	mean s/d
LoRA	93.12%	1,919,234	54.00	0.052	0.0089	0.0112	0.402
DoRA	92.55%	2,002,178	56.23	0.054	0.0093	0.0115	0.426
MAP	88.53%	1,919,378	5.36	0.001	0.0013	0.0018	0.002



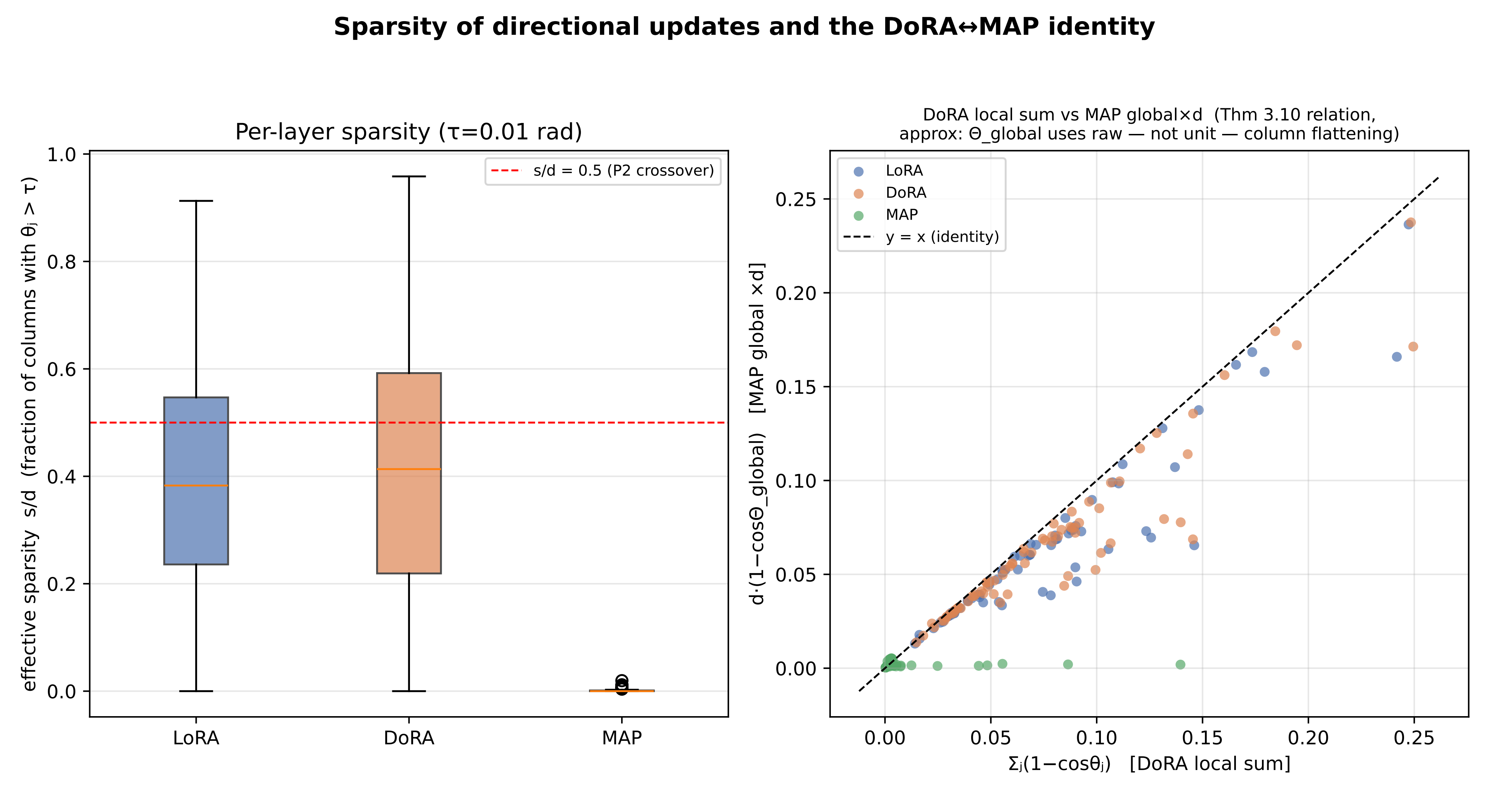


Θ_global is the single rotation of the whole flattened matrix (MAP view); θ_j are the per-column rotations (DoRA view).
Exact identity (Thm 3.10): $1-\cos\Theta\_global = mean\_j(1-\cos\theta_j)$ → DoRA sums what MAP averages.

Distribution of per-column directional change θ_j (pooled over all target layers)



A long right tail = a few columns rotate a lot (sparse/localized update). A concentrated body near 0 = most columns barely move.



Sparsity of directional updates and the DoRA↔MAP identity
Per-layer sparsity (τ=0.01 rad)
s/d = 0.5 (P2 crossover)
effective sparsity s/d (fraction of columns with θⱼ > τ)
1.0
0.8
0.6
0.4
0.2
0.0
LoRA
DoRA
MAP
DoRA local sum vs MAP global×d (Thm 3.10 relation,
approx: Θ_global uses raw — not unit — column flattening)
LoRA
DoRA
MAP
y = x (identity)
d·(1−cosΘ_global) [MAP global ×d]
0.25
0.20
0.15
0.10
0.05
0.00
Σⱼ(1−cosθⱼ) [DoRA local sum]

Where the rotation happens: attention vs MLP, and across depth